

Introduction to Biophysics

Technische Universität München

Wintersemester 2025/2026

Prof. Julia Herzen, Prof. Andreas Bausch, Prof. Ulrich Gerland

Solutions to Exercise Sheet 5

Problem 5.1

Problem 5.1a)

One can assume the shape of E.coli to be a cylinder or capsule.

- surface area of E. coli:

$$A_{\text{cylinder}} = \pi dh + 2\pi\left(\frac{d}{2}\right)^2 = \pi(2 \mu\text{m})(1 \mu\text{m}) + 2\pi\left(\frac{1}{4} \mu\text{m}^2\right) = 2.5\pi\mu\text{m}^2$$

$$A_{\text{capsule}} = \pi dh + 4\pi\left(\frac{d}{2}\right)^2 = \pi(1 \mu\text{m})(1 \mu\text{m}) + 4\pi\left(\frac{1}{4} \mu\text{m}^2\right) = 2\pi\mu\text{m}^2$$

- volume of E. coli:

$$V_{\text{cylinder}} = \pi\left(\frac{d}{2}\right)^2 h = \frac{\pi(1 \mu\text{m})^2(2 \mu\text{m})}{4} = \frac{\pi}{2} \mu\text{m}^3$$

$$V_{\text{capsule}} = \pi\left(\frac{d}{2}\right)^2 h + \frac{4}{3}\pi\left(\frac{d}{2}\right)^3 = \frac{\pi(1 \mu\text{m})^2(1 \mu\text{m})}{4} + \frac{4}{3}\pi\frac{(1 \mu\text{m})^3}{8} = \frac{5\pi}{12} \mu\text{m}^3$$

Problem 5.1b)

- mass of average bacterium: $m_{\text{bacterium}} = \rho V_{\text{bacterium}} = 1000 \frac{\text{kg}}{\text{m}^3} \cdot 1.5 \mu\text{m}^3 = 1.5 \cdot 10^{-12} \text{ g} = 1.5 \text{ pg}$
- number of bacteria: $N_{\text{bacteria}} = \frac{m_{\text{total}}}{m_{\text{bacterium}}} \approx \frac{200 \text{ g}}{1.5 \cdot 10^{-12} \text{ g}} = 1.3 \cdot 10^{14}$
- average mass of human cell: $m_{\text{human cell}} \approx 1 \text{ ng}$
- average number of human cells in a human: $N_{\text{human cells}} = \frac{70 \text{ kg}}{1 \text{ ng}} = 7 \cdot 10^{13}$

remark: more detailed estimates (considering organs individually) at around $N = 3.7 \cdot 10^{13}$, cf. doi:10.3109/03014460.2013.807878

Problem 5.1c)

Force required to sustain motion:

- force balance between force generated by E. coli and friction force:

$$F_{\text{E. coli}} = F_{\text{friction}} = \gamma v,$$

with friction coefficient γ and velocity v .

- friction coefficient γ as function of kinematic viscosity ν (via Stoke's law):

$$\gamma = 6\pi\nu\rho R,$$

with density of the medium ρ and size of the moving object (for simplicity assumed to be spherical) R

- combining both equations

$$\begin{aligned}F_{\text{E. coli}} &= 6\pi\nu\rho Rv = 6\pi \cdot \left(10^{-6}\frac{\text{m}^2}{\text{s}}\right) \cdot \left(1000\frac{\text{kg}}{\text{m}^3}\right) \cdot (1\ \mu\text{m}) \cdot \left(30\frac{\mu\text{m}}{\text{s}}\right) \\&= \left(1.88 \cdot 10^{-8}\frac{\text{kg}}{\text{s}}\right) \cdot \left(30\frac{\mu\text{m}}{\text{s}}\right) \\&= 5.65 \cdot 10^{-13}\ \text{N} = 0.57\ \text{pN}\end{aligned}$$

Distance travelled until stopping:

- friction force acting on E. coli:

$$m\frac{dv}{dt} = -\gamma v \rightarrow v(t) = v_0 \exp\left[-\frac{\gamma t}{m}\right]$$

note: velocity is zero for $t \rightarrow \infty$

- compute distance by integration of velocity:

$$x(t) = \int_0^t dt' v(t') = \int_0^t dt' v_0 \exp\left[-\frac{\gamma t'}{m}\right] = \frac{v_0 m}{\gamma} \left(1 - \exp\left[-\frac{\gamma t}{m}\right]\right)$$

- distance travelled until $v(t) = 0$, *i.e.*, distance travelled until $t \rightarrow \infty$:

$$x(t \rightarrow \infty) = \frac{v_0 m}{\gamma} = \frac{30\frac{\mu\text{m}}{\text{s}} \cdot 1.5\ \text{pg}}{1.88 \cdot 10^{-8}\frac{\text{kg}}{\text{s}}} = 2.4 \cdot 10^{-12}\ \text{m} = 2.4\ \text{pm}$$

- takeaway: E. coli stops almost instantaneously

Problem 5.2

Problem 5.2a)

- dimensionless time: $\tilde{t} = \mu t$

$\tilde{t}$ is dimensionless because $[\mu] = 1\frac{1}{\text{s}}$ and $[t] = 1\ \text{s}$

- implication: time measured in units of a characteristic growth time scale $\tau_{\text{growth}} = \frac{1}{\mu}$

- dimensionless number of bacteria: $\tilde{N} = \frac{N}{K}$

$\tilde{N}$ is dimensionless because $[N] = \text{"number of bacteria"}$ and $[K] = \text{"number of bacteria"}$

- dimensionless form of the differential equation:

$$\begin{aligned}\frac{dN}{dt} &= \mu N \left(1 - \frac{N}{K}\right) \\ \mu K \frac{d\tilde{N}}{d\tilde{t}} &= \mu K \tilde{N} (1 - \tilde{N}) \\ \frac{d\tilde{N}}{d\tilde{t}} &= \tilde{N} (1 - \tilde{N}) \\ \frac{d\tilde{N}}{d\tilde{t}} &= f(\tilde{N})\end{aligned}$$

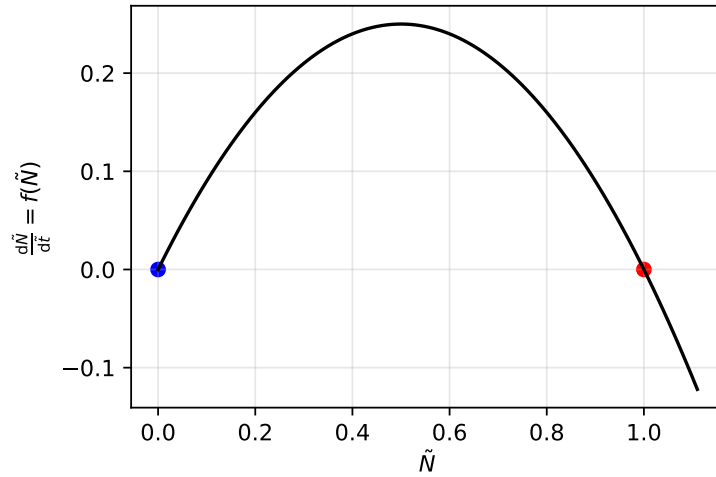


Figure 1: Plot of $f(\tilde{N})$. The fixpoints are highlighted in green and red. The green fixpoint is (linearly) unstable, the red fixpoint (linearly) stable.

Problem 5.2b)

number of bacteria for which maximal growth is attained: maximum of growth rate implies that $f(N)$ is maximal

$$\frac{\partial f(\tilde{N})}{\partial \tilde{N}} = \frac{\partial}{\partial \tilde{N}} \tilde{N}(1 - \tilde{N}) = 1 - 2\tilde{N} = 0 \rightarrow \tilde{N}^* = \frac{1}{2}$$

Problem 5.2c)

maximum growth rate:

$$f(\tilde{N}^*) = \frac{1}{2} \cdot \left(1 - \frac{1}{2}\right) = \frac{1}{4}$$

Problem 5.2d)

- plot of $f(\tilde{N})$ as a function of $\tilde{N}$, cf. Figure 1: quadratic function with maximum at $\tilde{N} = \frac{1}{2}$, note that only positive arguments $\tilde{N} \geq 0$ are valid as number of bacteria cannot be negative
- definition of fixpoints: At a fixpoint, the observable does not change in time, *i.e.* $\frac{d\tilde{N}}{dt} = 0$. Thus, all fixpoints obey $f(\tilde{N}) = 0$.
- stability of green fixpoint at $\tilde{N}_g = 0$: If the system is slightly perturbed away from the fixpoint (*i.e.*, $\tilde{N}$ is slightly increased), $\frac{d\tilde{N}}{dt}$ becomes positive. Therefore, $\tilde{N}$ increases even further and is driven away from the fixpoint. The fixpoint is unstable.

more formal reasoning: consider a Taylor expansion around $\tilde{N}_g = 0$,

$$\begin{aligned} \tilde{N} &= \tilde{N}_g + \epsilon = \epsilon \\ \frac{d\epsilon}{dt} &= \epsilon(1 - \epsilon) = \epsilon + \mathcal{O}(\epsilon^2) \\ \rightarrow \epsilon(\tilde{t}) &= \epsilon_0 e^{\tilde{t}} \end{aligned}$$

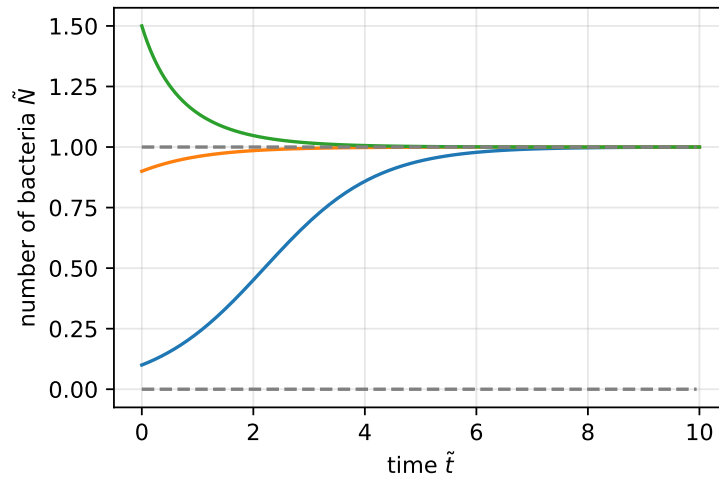


Figure 2: Time evolution of $\tilde{N}(t)$ for different initial conditions $\tilde{N}(t = 0)$ obtained via numerical integration of the differential equation. For any initial condition, the system converges towards the stable fixpoint $\tilde{N}_r$.

Any small perturbation increases exponentially in the short time limit.

- stability of the red fixpoint at $\tilde{N}_r = 1$: If the system is perturbed away from the fixpoint by increasing $\tilde{N}$, $\frac{d\tilde{N}}{dt}$ becomes negative and forces $\tilde{N}$ back towards $\tilde{N}_r$. The same is the case if the system is perturbed away from the fixpoint by decreasing $\tilde{N}$, as then $\frac{d\tilde{N}}{dt}$ becomes positive and increases $\tilde{N}$ again.

more formal reasoning: consider a Taylor expansion around $\tilde{N}_r = 1$,

$$\begin{aligned}\tilde{N} &= \tilde{N}_r + \epsilon = 1 + \epsilon \\ \frac{d\tilde{N}}{dt} &= \frac{d\epsilon}{dt} = (1 + \epsilon)(1 - 1 - \epsilon) = -\epsilon + \mathcal{O}(\epsilon^2) \\ \rightarrow \epsilon(t) &= \epsilon_0 e^{-t}\end{aligned}$$

Any small perturbation relaxes exponentially back towards the fixpoint.

- Add-On: Numerical integration of the differential equation shows that the system converges towards the fixpoint $\tilde{N}_r = 1$ irrespective of the starting conditions, i.e., this fixpoint is not only linearly stable, but even a global fixpoint of the system, cf. Figure 2.

Problem 5.3

Main observation of the Paper

The authors examine bacterial cultures (of the species *Salmonella typhimurium*) at different bacterial growth rates μ . They measure the concentration R of different ribosomal species (the so-called 35S, 55S, 70S and 85S sedimentation fractions) and compare them to the concentration P of the soluble proteins (7S fraction), to obtain the ratio R/P . They find that this ratio grows approximately proportional to μ .

Problem 5.3a)

The cells are growing at a rate μ , which implies that the number N of cells evolves according to

$$\frac{dN}{dt} = \mu N .$$

This means, in turn, that the total number of proteins is given by, $dA/dt = \mu A$, since $A = NA_{\text{cell}}$, where A_{cell} is the amount of protein per cell. By assumption that total protein production is proportional to the number of ribosomes and hence $dA/dt = yR$. This implies

$$\mu A = yR$$

and thus

$$\frac{R}{A} = \frac{\mu}{y} .$$

This suggests that $\mu = y$ is the maximum growth rate, since in this case we have the condition $R = A$ meaning that effectively all protein production goes to making new ribosomes.

Problem 5.3b)

A cell cycle of 3000 s corresponds to $\mu = 0.8/\text{hr}$ (50 min = 1.2 doublings/hr = 0.8 e-fold growth/hr). As a result we get

$$\frac{R}{A} = 0.25 = \frac{\mu}{y} ,$$

which leads to $y = 3.2/\text{hr}$. Note that here we've converted from base 2 (doubling times) to base e, since for the growth rate equation, $dP/dt = \mu P$, the solution is proportional to $e^{\mu t}$, indicating that $1/\mu$ actually represents the time to go from a starting P to e times this. For the purposes of the answers to this question this doesn't matter, since the additional factor picked up by doing this drops out in the ratios.

With 2.5 doublings/hr (the maximum in the graphs in figure 2 of the paper) = 1.7 e-fold/hr, we have $\mu = 1.7/\text{hr}$, and

$$\frac{R}{A} = \frac{1.7}{3.2} ,$$

which leads to $R/A \approx 0.5$, meaning that half of the total protein content of the cell is given by ribosomes. Now, we want to compare that to the magnitude measured in their paper, R/P . Assuming that the total protein content is just the sum of ribosomal protein plus soluble protein, $A = R + P$, we get

$$\frac{R}{P} = \frac{R}{A - R} = \frac{1}{\frac{A}{R} - 1} .$$

As a result, for the ratio $R/A \approx 0.5$ calculated in the simple model we get $R/P = 1$. This number is a clear overestimate with respect to the actual measurements, where they measured this ratio to be $R/P \approx 0.6$.

Using the linear model from the paper given by $\mu = kR/P$ and combining this with the above equation for R/P , we get

$$k = \mu \left(\frac{A}{R} - 1 \right) = 0.8 \text{ hr}^{-1} \times 3 = 2.4 \text{ hr}^{-1} .$$

For the highest growth rate, $\mu = 1.7/\text{hr}$, we then get

$$\frac{R}{A} = \left(\frac{k}{\mu} + 1 \right)^{-1} = \left(\frac{2.4 \text{ hr}^{-1}}{1.7 \text{ hr}^{-1}} + 1 \right)^{-1} \approx 0.4$$

$$\frac{R}{P} = \frac{\mu}{k} = \frac{1.7 \text{ hr}^{-1}}{2.4 \text{ hr}^{-1}} \approx 0.7$$

which leads to $R/A \approx 0.4$, or $R/P \approx 0.7$, much closer to the actual value measured!